WHY IT COLLAPSED

HOW A LARGE PART OF THE UNDER-CONSTRUCTION AGUWANI-SULTANGANJ BRIDGE PLUNGED INTO THE RIVER GANGA TWICE IN 14 MONTHS

Graphic by **NILANJAN DAS** | *Text by* **AMITABH SRIVASTAVA**

FIRST COLLAPSE
APRIL 30, 2022

- Pylon 5, involving 162.5 metre of the extradosed span of the 3,160 metre bridge, crumbles
- A total of 19 segments fall into the river while two segments on both sides are damaged
- In March 2023, the state-owned Bihar Rajya Pul Nirman Nigam Ltd (BRPNNL) ensured dismantling of six segments on both sides of Pylon 5 and ordered removal of the pylon and pier table at the cost of the contractor. This was done in accordance with an IIT Roorkee investigation report

Pylon: *A tower that gives support*

Span: *The distance between two intermediate supports for a structure*

Extradosed: *A cross between a girder bridge and a cable-stayed bridge*

SECOND COLLAPSE
JUNE 4, 2023

- Pylon 9, 10, 11, 12 and 13 of the extradosed module collapse. Total length: 575 metres

₹100 CRORE
ESTIMATED LOSS



WHAT WENT WRONG

- The root cause of the collapse, as per the IIT Roorkee report, was the failure of the match casting on site. As the segments of the under-construction bridge failed to fit perfectly with each other and transfer the load across the entire surface, the span's profile kept going out of bounds
- Also, concrete parameters as assumed in design are not achieved on site. The quality of wet stitches were not up to the design strength
- BRPNNL officials believe the flaws that led to the collapse of Pylon 5 in April 2022 are also the reason for the second collapse in June
- The onus of the structural flaws in the under-construction bridge segment lies squarely with S.P. Singla Construction Pvt. Ltd, the contractors. Also, the BRPNNL's executive engineer was found wanting in his role—he should have monitored the construction process closely and intervened